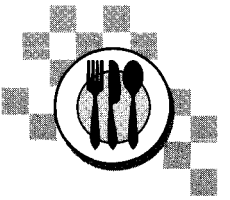
CHAPTER 18



Foodservice Control Points

About This Chapter

In this chapter you will learn:

1. How to keep food safe at each step in a foodservice facility, from buying and receiving to serving leftovers
2. How to buy, receive, and store food safely
3. How to prepare, serve, cool, and reheat safely
4. How to draw flow diagrams for each recipe used in the facility to help decide on critical control points (CCPs)
5. How to monitor CCPs to make sure each batch of food is safe
6. How the design of the facility and its equipment affect food safety

Introduction

In a foodservice operation, the flow of food begins when it is bought and delivered (purchasing and receiving) and continues through storing, preparing, cooking, holding, and serving. In a foodservice facility, the sequence may include these steps:

- Buying ingredients
- Receiving ingredients when the supplier delivers them
- Storing ingredients
- Preparing ingredients (thawing, processing)
- Cooking
- Holding or displaying food
- Serving food

- Cooling and storing leftover food
- Reheating and serving food

Each step provides opportunities for food to be contaminated or for bacteria to grow. Food operators need to look at each step to find out what the hazards are (hazard analysis) and how they can make sure they are under control (critical control points, CCPs).

This chapter will discuss how foodservice operators can use control points as food flows through the system to make sure that they always serve safe food to customers.

BUYING INGREDIENTS

It is important to buy foods from reliable suppliers who meet federal and local standards. Managers should be sure that suppliers understand their specifications (detailed descriptions) of the food they want.

Meat and Poultry

All meat and poultry should have a stamp on the carcass, parts, or packaging material to show that the USDA or state department of agriculture has inspected it. Figure 18.1 shows inspection stamps for meats and poultry. Suppliers should be able to show written proof that the government has inspected the meat or poultry.

Shellfish

Shellfish suppliers should be on public-health-service Food and Drug Administration lists of Certified Shellfish Shippers or on lists of state-approved sources. Managers must keep shellstock identification tags on file for 90 days after receiving clams, mussels, and oysters.

RECEIVING AND STORING INGREDIENTS

Employees must check and store food as quickly and efficiently as possible. They must know how to decide whether each food is acceptable and how to take tempera-



FIGURE 18.1. Inspection stamps for meat and poulry: left, inspection stamp for meat; right, inspection stamp for poultry. (Source: U.S. Department of Agriculture.)

tures of refrigerated and frozen foods. Storage areas must be large enough to store foods safely, cleanly, and at the right temperature. Dry storage areas must be clean and dry. Be sure that employees:

- Do not store food in restrooms, furnace rooms, or hallways.
- Do not stack foods on floors.
- Put food on shelves with enough space around them for air to circulate.
- Wrap food in waterproof, nonabsorbing, clean coverings.
- Do not put food directly onto shelves.
- Store above raw foods all cooked foods and foods that will not be cooked any more.
- Follow the "First In, First Out" (FIFO) rule for rotating stock (i.e., use the oldest food first).

Fresh Meat, Poultry, Seafood, and Shellfish

Staff must check the temperature of fresh meat, poultry, seafood, and shellfish when it arrives. Managers should train employees to judge the freshness of the food based on color, texture, smell, and general appearance. Cartons and wrappers must be clean and unbroken.

Frozen foods

All frozen foods, except ice cream, should be –18°C (0°F) or colder when the supplier delivers them. Ice cream may be –14 to –12°C (6°–10°F). An assigned employee should check the temperature by opening one case and putting the sensing area of the thermometer between two packages without puncturing them. A trained employee should look for thawed and refrozen food that has large ice crystals, solid areas of ice, or a lot of ice inside the container. Dried-out food or food that has changed color should be rejected. The packages should be airtight and waterproof.

Canned and Dry Goods

Foodservice facilities should never accept home-canned foods because of the risk of botulism. It is essential to check that cans are not swollen, rusty, or dented, and do not have flaws in the seals or seams. The establishment should return cans if the contents are foamy or smell bad. Employees should never taste the contents to test them, because tiny amounts of bacterial food poison could make them very ill.

Dairy Products

Milk must be pasteurized and should be delivered well before its expiration date. Other milk products, such as cream, dried milk, cottage cheese, soft cheese, and cream cheese, must be made from pasteurized milk. Dairy products should not be sour or moldy.

Employees should check the temperature of milk cartons by opening one and putting in a sanitized thermometer. Employees can check the temperature of bulk plastic containers by folding the soft plastic around the thermometer without puncturing the package. For other products, they should put the thermometer between two packages. Dairy products should be 4°C (40°F) or below.

Eggs and Egg Products

Eggs should be grade AA or A and should have the USDA shield on the carton. Eggs should be refrigerated and less than 2 weeks old. Liquid eggs should be refrigerated or frozen, even though they are pasteurized. Egg shells must be clean and uncracked.

Refrigerated Prepared Foods

Many establishments now buy foods that are partly or completely processed and prepared and can be served to customers with little or no preparation. These foods include refrigerated entrees, prepared salads, fresh pasta, soups, sauces and gravies, and cooked or partly cured meats and poultry dishes. They are sometimes called fresh-prepared foods, fresh-refrigerated foods, or ready-to-eat chilled foods. The foodservice operator has no way to control the safety of these foods (except to store them at the right temperature and use them within the right amount of time). Therefore, the operator should make sure the supplier has a good HACCP program.

If the package has a time-temperature indicator, employees should check it to make sure the food has been stored properly. The right storage temperature is listed on the label of the food.

Foods with Modified Atmosphere Packaging

Modified atmosphere packaging (MAP) is usually used for partially processed foods that are cooked lightly first and then packaged. MAP contains a mix of gases inside the package to lengthen the shelf life of the food. These products may be shelf-stable (heat sterilized: require no refrigeration before they are opened), refrigerated, or frozen. The package may be a pouch, tray, plastic bucket, or glass bottle.

Employees should check the expiration date and/or time-temperature indicator of MAP foods when they are delivered. The gases in the package stop spoilage bacteria from growing, and so the food may not look spoiled but may still have pathogens growing inside. Therefore, it is important to store MAP foods at the right temperature and only until the expiration date.

Sous Vide Foods

Sous vide is a French term that means "under vacuum." Manufacturers put the food in a strong plastic pouch, remove all the oxygen and other air gases, and seal it. Foodservice operators are not allowed to make *sous vide* foods on site, but they can buy them from licensed food processors. *Sous vide* foods may by frozen, refrigerated,

or cooked, cooled, and frozen or refrigerated. Uncooked foods are cooked in the sealed pouch by putting it in boiling water.

Frozen *sous vide* foods should be –18°C (0°F) or colder when they arrive. Refrigerated *sous vide* foods should be about 1°C (33–34°F). Higher temperatures shorten the shelf life of the food and may make it unsafe. Again, employees should check temperatures, time-temperature indicators, and expiration dates. Also, they must make sure the package seal is not broken.

Aseptic and Ultrapasteurized Packaged Foods

Aseptic packaged foods are heat-treated to kill pathogens. Foods labeled UHT have been ultrapasteurized (high temperature for a short time) and packaged in a sterile environment. Many UHT foods do not need to be refrigerated, but UHT milk should be 4°C (40°F) or colder.

PREPARING, SERVING, AND REHEATING FOODS

Deciding on CCPs

Most CCPs are in the preparing, serving, and reheating steps. The best way to find the CCPs is to draw a flowchart for each recipe. The flowchart begins with each ingredient and ends with the finished food product or leftovers. Every time the foodservice operation adds a new menu item, managers should draw a flowchart and look for the CCPs. Management should do this when they develop the recipe.

Figure 18.2 shows a recipe for chili. The flowchart in Figure 18.3 shows the flow of food from the time a supplier delivers the ingredients until a customer eats the leftovers. CCPs are shaded. CCPs are any step that needs to be controlled and checked to make sure the food is safe and sanitary. CCPs are different for each food but may include:

- Buying foods from reliable suppliers
- Cooling food within a set amount of time
- Storing food below 4°C (40°F) in the refrigerator or below –18°C (0°F) in the freezer
- Cooking food until the inside reaches a set temperature
- Cooking food for a set amount of time at a set temperature
- Keeping food hotter than 60°C (140°F) or cooler than 4°C (40°F) while it is served
- Not mixing old and new batches of food
- Reheating foods to 74°C (165°F) for a set amount of time

Checking and Monitoring CCPs

It is not enough to decide on the CCPs. Management needs to make sure that all food production meets CCP standards. Managers should set up a record-keeping system. They should design forms that list the CCPs for each recipe, have spaces to record temperatures and times, and have places to log when a CCP is carried out. Employees should complete the forms for every batch of food they prepare.

INGREDIENTS	AMOUNTS
Ground beef	4.5 kg (10 lb)
Onions	2.25g (8 oz)
Tomatoes, canned, diced	2.5L (2 1/2 qt)
Tomato purée	2L (2 qt)
Chili powder	56g (2 oz)
Salt	56g (2 oz)
Pepper, black	1/2 t
Beans, kidney or red, canned	4.25kg (9 lb 8 oz)
Water	1L (1 qt)

PREPARING

1. Chop onions.
2. Drain and rinse beans.

COOKING

3. *Brown beef in stock pot until it reaches 68°C (155°F) for at least 15 seconds.* Drain and add onions.
4. Add tomatoes, tomato purée, chili powder, salt, pepper, beans, and water to stockpot.
5. Heat, stir, bring to boil, then cover and simmer 1 1/2 to 2 hours.

SERVING AND HOLDING

6. Serve immediately.
7. *Hold chili at 60°C (140°F) or higher.* Do not mix new product with old.

COOLING

8. Cool leftovers in shallow pans so that the layer of chili is no more than 2 inches deep. *Cool to 4°C (40°F) or lower within 6 hours.* Stir often.
9. Cover and store at 4°C (40°F) or lower in refrigerator.

REHEATING

10. *Reheat chili to at least 74°C (165°F) for at least 15 seconds within 2 hours. Reheat one time only. Discard any unused food after serving it the second time.*

Note: Italics indicate critical control points (CCPs)

FIGURE 18.2. Chili recipe.

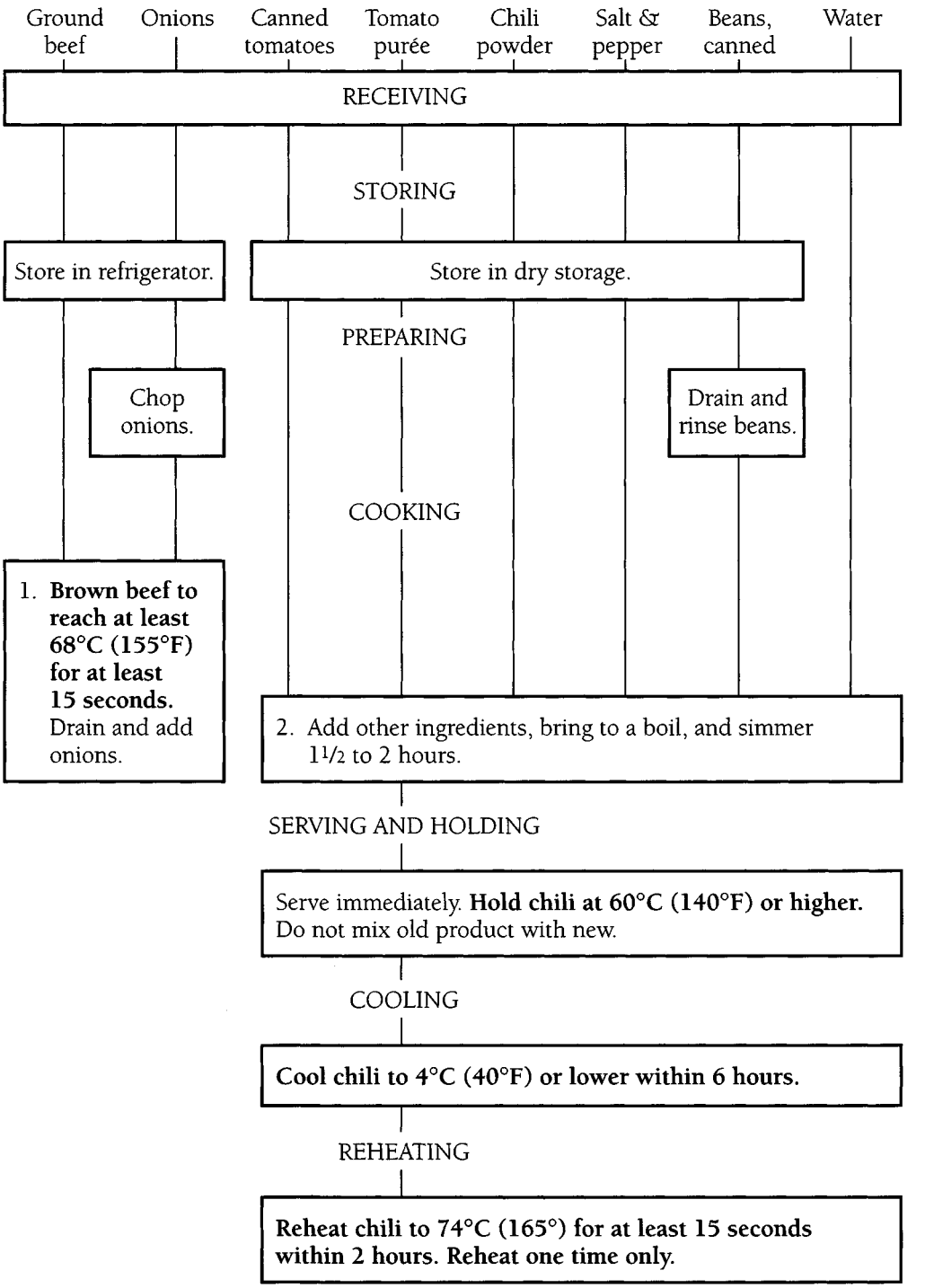


FIGURE 18.3. Chili flowchart. Boldface statements are critical control points (CCPs).

Employees need the right equipment to make sure that they meet CCPs, such as:

- Accurate thermometers; managers should calibrate thermometers at least once a day
- Clocks and timers in convenient locations
- Proper facilities for washing hands, utensils, and equipment
- Enough refrigerator and freezer storage space
- Ice and shallow containers for chilling foods

Standard Procedures

A foodservice operation needs more than just CCPs to make sure that the food is safe. Standard procedures for preparing, cooking, cooling, and reheating foods protect foods from contamination and conditions that would allow pathogens to grow. *All foods that could allow pathogens to grow should be prepared so that they remain less than 4 hours total in the temperature danger zone (4–60°C, 40°–140°F) from the time the ingredients are taken out of storage until the food is served to the customer.* Ways to make sure foods do not spend too much time in the danger zone are:

- Keep refrigerated foods in the refrigerator right until they are prepared.
- Prepare foods as soon as possible before serving them.
- Prepare small batches and keep them in the refrigerator between preparation steps.
- Cool cooked foods quickly by placing shallow pans in ice water.
- Reheat foods quickly and keep them hot during service.
- Serve foods on steam tables or ice beds.

Thawing. Thawing may take several hours or days for large frozen items. It is important to thaw foods safely so that the outside is not warm enough to allow microbes to grow while the inside is still frozen. Choose one of these methods to thaw foods:

- Refrigerate at 3°C (38°F) or colder. Store raw foods on the lowest shelves so that they do not drip or splash onto other foods. Large items, such as turkeys or roasts, take a day or more to thaw (allow about 1 day for each 4–5 lb).
- Cook frozen food to the proper internal temperature (see Appendix 3). This works for vegetables, seafood, hamburger patties, and pie shells, but not for large items, such as turkeys. The cooking time is about 50% longer if the food is frozen (i.e., if it would usually take 1 hour to cook, allow 1.5 hours).
- Use a microwave only if the food will be cooked immediately after it is thawed. This method does not work well for large items, such as turkeys.

Cooking. It is important to use a thermometer to check the temperature inside the food at the end of the cooking time for each batch of food. Employees should not decide that the food is cooked based on how it looks or how long it has been cooking. Other important cooking procedures are:

- Do not overload ovens or cooking surfaces.
- Make sure each portion is the same size and thickness and has the same fat content, so that each one cooks at the same speed.
- Make sure the oven or stove gets back to the right temperature between cooking batches of food.
- Do not touch cooked food with bare hands. Use sanitized serving utensils and cutting boards.

Holding and displaying. Holding equipment should keep hot foods hotter than 60°C (140°F) and cold foods cooler than 4°C (40°F). Other important points are:

- Make sure food is 73°C (165°F) or hotter before transferring it to hot-holding equipment, such as a steam table, double boiler, *bain-marie,* or chafing dish. Do not use holding equipment to cook or reheat food, only to keep it hot.
- Never add new product to old food. Replace the old dish of food with a new dish of food, and replace the serving utensil at the same time.
- Measure temperatures every 2 hours.
- Make sure ice around chilled foods drains away from food containers.
- Stir hot foods often and protect them with lids and sneeze guards.

Cooling and storing. Cooling food the wrong way is the most common cause of foodborne illness. Proper cooling takes time and effort. The temperature inside the food should be below 4°C (40°F) within 4 hours after cooking or the end of hot holding. FDA's 1995 Food Code states that food must cool to 21°C (70°F) within 2 hours, and then to 5°C (41°F) within 4 more hours (total = 6 hours). Cooling needs good planning.

- Use a quick-chill unit, if available, for large food items divided into smaller quantities (such as turkey and roasts), or hot, thick items (such a stews and casseroles).
- Cut large items into pieces. Cut the pieces small enough to cool fast enough. You may need to test how quickly different-size pieces of food cool.
- Use an ice bath or cold-jacketed kettles. Stir or shake foods often while they cool.
- Put the food in shallow pans on the top shelves of a refrigerator. Prechill pans, and use pans made of stainless steel or other materials that transfer heat quickly. Pans should not be more than 10 cm (4 inches) deep, and food should not be more than 5 cm (2 to 3 inches) deep. Do not stack pans, because air cannot circulate around stacked pans.
- Measure the temperature of the food to make sure it is cooling quickly enough. If the food has not cooled within 4 hours, reheat it to 74°F (165°C) or higher for at least 15 seconds within 2 hours, and recool, or throw it out.
- Label foods with the date and time they were prepared, cover, and store.

Reheating. It is important to reheat food thoroughly to 74°C (165°F) within 2 hours of taking it out of the refrigerator so that pathogens in the food are killed and do not have time to grow. Employees must:

- Not reheat food more than once.
- Keep leftover and fresh foods separate.
- Only reheat foods that have been cooled and refrigerated for 2 days or less.
- Never reheat food in hot-holding equipment.

FACILITY AND EQUIPMENT DESIGN

It is hard to maintain good sanitation in a badly designed foodservice facility. Cleaning tasks are difficult and frustrating, and staff waste time and energy trying to clean properly. This often makes workers give up on complete and thorough cleaning.

Cleaning the Facility Easily

Employees must be able to clean and inspect equipment and surfaces easily using normal cleaning methods. Soil, pests, and microorganisms should not have places where they can collect and live. People will be more motivated to clean and keep the facility sanitary if it takes less effort.

Designing the Facility

Operators should consider sanitation when they plan the facility. Most managers work in established foodservice facilities, but they can improve the design if the operator remodels, renovates, or buys new equipment.

Most areas have laws that make sure that foodservice units have a sanitary design. Public-health, building, and zoning departments can all make regulations about the construction of foodservice operations. Regulatory agencies often have checklists to help operators make sure that the building design is sanitary.

Floors, walls, and *ceilings* should be easy to clean and maintain and should also look clean. Operators should use materials that do not react with food or cleaning chemicals, last a long time, do not absorb dirt, and are smooth. If the floor, wall, or ceiling absorbs liquid, the liquid can damage the surface, microbes can grow on it, and it will develop an offensive odor. Food preparation and food storage areas should not have carpets, rugs, or other porous coverings.

The way the floor is built is also important. Coving at the joint between the floor and the wall makes cleaning easier and makes it less likely that bits of food will be left to attract insects and rodents. Operators should seal concrete and terrazzo floors so that they do not absorb liquid or create cement dust.

For walls, ceramic coverings are popular and work well in most areas. Grouting should be smooth and waterproof, and should not have cracks or holes that could collect soil. Stainless steel is expensive but is a good finish because it keeps out water and most soil and lasts a long time. Plaster painted with nontoxic paint or cinder block walls are fine in dry areas, as long as the builder seals them with soil-resistant and glossy paints, epoxy, acrylic enamel, or another good seal. Foodservice operators should not use lead-based or other toxic paints because flakes or chips may get into the food. Ceilings should have smooth surfaces that do not absorb water and are

easy to clean. Smooth-sealed plaster, plastic panels, or plastic-coated panels are all good choices.

When the foodservice manager buys new equipment, he or she can make sure that it meets good sanitation standards. Equipment should:

- Have as few parts as possible to do the job
- Be easy to take apart
- Have smooth surfaces with no pits, crevices, ledges, bolts, and rivet heads
- Have rounded edges and corners inside with smooth surfaces
- Be coated with materials that do not crack or chip
- Be made of materials that are not toxic, do not absorb liquids or fats, and do not color or flavor the food

Installing and Laying Out Equipment

Operators should arrange equipment so that food is not easily contaminated and all areas are easy to reach and clean. For example, the dirty-dish table should not be put next to the vegetable preparation sink. Waste and food preparation areas should be as far apart as possible, and food preparation equipment should not be put under an open stairway.

Mobile equipment is best whenever possible, because with such equipment, staff can easily clean the walls and floors around it. Operators should seal equipment that is not mobile to the wall or to other equipment. If they cannot seal the equipment, they should put it 0.5 m (1.5 ft) away from the wall or other equipment so that staff can get around it to clean. Operators should mount equipment that is not mobile about 0.25 m (9 in) off the floor or seal it to a masonry base.

If operators use a seal, it should not be toxic. Operators should not use seals to cover gaps from bad construction, because after a while the cracks will open up and collect soil, insects, and rodents.

Handwashing Facilities

Hands are the most common source of microbes. Therefore, management should provide places for employees to wash their hands wherever hands are likely to get contaminated: in food preparation areas, lockerrooms or dressing rooms, and next to toilets. Employees may not want to walk very far to wash their hands, so handwashing sinks need to be close to where they work. Handwashing sinks should have a washbowl, hot and cold water with foot-operated faucets, liquid or powdered soap, and individual towels or air dryers.

Dressing and Lockerrooms

Employees should have dressing rooms or lockerrooms. Street clothes can be contaminated with microbes, so the operator should provide uniforms for employees to wear while they work. The facility should have a wall or other barrier between dress-

ing rooms and food preparation, storage, and serving areas. Handwashing sinks should be next to the dressing rooms and toilet rooms. Handwashing sinks should not have mirrors above them, so that workers are not tempted to touch their faces or hair after they wash their hands. Staff should scrub the washing facilities and toilet rooms and empty trashcans at least once a day.

Waste Disposal

Garbage and trash disposal need to be done properly, because waste products can attract pests that contaminate food, equipment, and utensils. Trashcans should not leak, should keep out pests, should be easy to clean, and should last a long time. Plastic bags and wet-strength paper bags are good liners for trashcans. Cans must have tight-fitting lids.

Staff should remove trash from food preparation areas often so that it does not start to smell bad or attract pests. Staff should only put waste materials in trashcans. Waste storage areas should be easy to clean and free of pests. If the facility has to keep trash for a long time, the storage area should be indoors and refrigerated. Large dumpsters and compactors outside the building should be on or above a smooth surface that does not absorb liquids like concrete or machine-laid asphalt.

Trash areas should have a supply of hot and cold water and a drain. It is important that water does not contaminate the food preparation or storage areas when staff clean the trash area.

Operators can reduce the amount of trash using a pulper or mechanical compactor. Pulpers grind waste material into pieces that are small enough to flush away with water. The waste is dried, and the processed solid wastes can be trucked away. Compactors squash trash so that it only takes up about 20% of the original amount of space. Compactors are useful in facilities that do not have much space to store trash.

Operators may be able to incinerate (burn) some trash if it is allowed in their area and the incinerator meets federal and local clean-air standards. Most waste from foodservice establishments has too much moisture to burn well.

SUMMARY

- In a foodservice operation, the flow of food has several steps, including buying, receiving, storing, preparing, cooking, holding, serving, cooling, and reheating.
- Operators should buy food from reliable suppliers. Employees should check that the food is fresh, packages are clean and not damaged, and chilled or frozen foods are cool enough when they are delivered.
- The facility must have enough dry storage, refrigerator, and freezer space to stack foods off the ground and with enough space for air to circulate.
- Staff should use the FIFO principle (first in, first out) and should use older food first.
- The flow of food for each recipe can be drawn as a flowchart. This helps identify CCPs.

- Facilities need to monitor CCPs and use good standard procedures to keep food safe.
- The design of the facility and the equipment need to support safe food preparation.

BIBLIOGRAPHY

Bryan, F. L. Hazard analysis critical control point (HACCP) systems for retail food and restaurant operations. *J. Food Protection* 53(11):978–983.

The Educational Foundation of the National Restaurant Association. 1992. *Applied Foodservice Sanitation,* 4th ed. Kendall/Hunt, Dubuque, Iowa.

The Educational Foundation of the National Restaurant Association. 1993. *HACCP Reference Book.* The Educational Foundation of the National Restaurant Association: Chicago.

The Educational Foundation of the National Restaurant Association. 1995. *Serving Safe Food: A Practical Approach to Food Safety.* Kendall/Hunt: Dubuque, Iowa.

U.S. Public Health Service. 1978. *Food Service Sanitation Manual.* U.S. Government Printing Office, Washington, DC.

STUDY QUESTIONS

1. List six of the steps in the flow of food through a foodservice facility. Which step most often causes foodborne illness if it is not done properly?
2. What temperatures should refrigerated and frozen foods be kept at?
3. How quickly must food be cooled to refrigerator temperatures?
4. Foods must be reheated to at least _____ (temperature) for at least _____ (time).
5. What are three safe ways to thaw food?
6. What are three important things about handwashing sinks?

TO FIND OUT MORE ABOUT FOODSERVICE CONTROL POINTS

1. Draw a flowchart for a recipe. Use a foodservice recipe from your workplace or from a textbook. Highlight CCPs for the recipe. Does the recipe need to be prepared differently to make sure the food is safe?
2. Contact the Educational Foundation of the National Restaurant Association, 250 South Wacker Drive, Suite 1400, Chicago, IL 60606-5834, (1-800) 765-2122, for more information on HACCP in foodservice.
3. Use a foodservice-type food thermometer (thermocouple) to practice checking the internal temperatures of food at home.